Topology MATM36: Course Summary of Important Results

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Metric Spaces

Definition 1. *Let X be a topological space and $A \subseteq X$. A point $x \in X$ is called an* adherent point *of A if every open neighbourhood U of x satisfies*

$$U \cap A \neq \varnothing.$$

In the special case where X is a metric space with metric d, this is equivalent to requiring that for every $r > 0$,

$$B_r(x) \cap A \neq \varnothing.$$

Definition 2. *Let X be a topological space and $A \subseteq X$. The set A is called* dense *in X if every nonempty open set $U \subseteq X$ satisfies*

$$U \cap A \neq \varnothing.$$

Equivalently, A is dense in X if $\overline{A} = X$.

Definition 3. *Let X be a topological space and $A \subseteq X$. The set A is called* nowhere dense *in X if*

$$\operatorname{int}(\overline{A}) = \varnothing.$$

Definition 4. *A topological space X is* separable *if: $\exists D \subseteq X : D$ countable and dense.*

Baire Category Theorem

Theorem 1 (Baire Category Theorem)**.** *Let (X, d) be a complete metric space and let $\{U_n\}_{n=1}^{\infty}$ be a sequence of dense open subsets of X. Then $\bigcap_{n=1}^{\infty} U_n$ is dense in X.*

Proof. Fix a nonempty open ball $B(x, \varepsilon) \subset X$.

Since U_1 is dense and open, choose $y_1 \in U_1 \cap B(x, \varepsilon)$ and $r_1 > 0$ such that

$$\overline{B}(y_1, r_1) \subset U_1 \cap B(x, \varepsilon), \qquad r_1 < 1.$$

Inductively, having chosen y_n, r_n, use that U_{n+1} is dense and open to pick $y_{n+1} \in U_{n+1} \cap B(y_n, r_n)$ and $r_{n+1} > 0$ such that

$$\overline{B}(y_{n+1}, r_{n+1}) \subset U_{n+1} \cap B(y_n, r_n), \qquad r_{n+1} < \frac{1}{n+1}.$$

Then the closed balls are nested:

$$\overline{B}(y_{n+1}, r_{n+1}) \subset \overline{B}(y_n, r_n) \subset \cdots \subset \overline{B}(y_1, r_1) \subset B(x, \varepsilon),$$

and for $m > n$ we have $y_m \in B(y_n, r_n)$, hence $d(y_m, y_n) < r_n \to 0$. Thus (y_n) is Cauchy, so by completeness $y_n \to y \in X$.

For each fixed n, all tails of the sequence lie in $\overline{B}(y_n, r_n)$, so the limit satisfies $y \in \overline{B}(y_n, r_n) \subset U_n$. Hence $y \in \bigcap_{n=1}^{\infty} U_n$, and also $y \in \overline{B}(y_1, r_1) \subset B(x, \varepsilon)$. Therefore

$$B(x, \varepsilon) \cap \bigcap_{n=1}^{\infty} U_n \neq \varnothing,$$

so the intersection is dense. □

Remark 1. *From the definitions it follows that a set $B \subseteq X$ satisfies* $\operatorname{int}(B) = \varnothing \iff X \setminus B$ *is dense in X. In particular, if A is nowhere dense, then $X \setminus \overline{A}$ is open and dense in X.*

Corollary 1. *Let (X, d) be a complete metric space and let $\{F_n\}_{n=1}^{\infty}$ be a sequence of nowhere dense subsets of X. Then*

$$\operatorname{int}\left(\bigcup_{n=1}^{\infty} F_n\right) = \varnothing.$$

Proof. Since F_n is nowhere dense, $U_n := X \setminus \overline{F_n}$ is open and dense; by Baire, $\bigcap_{n=1}^{\infty} U_n = \bigcup_{n=1}^{\infty} \overline{F_n}$ is dense, hence nonempty in every nonempty open set. □

Finite Products of Metric Spaces

Definition 5. *Let (X_k, d_k), $k = 1, \ldots, n$, be metric spaces and set*

$$X = X_1 \times \cdots \times X_n.$$

A metric d on X is said to be a product metric *if convergence in (X, d) is coordinatewise,*

$$x^{(m)} = (x_1^{(m)}, \ldots, x_n^{(m)}) \to x = (x_1, \ldots, x_n)$$

in X if and only if for each $k = 1, \ldots, n$,

$$x_k^{(m)} \to x_k \quad \text{in } X_k.$$

Theorem 2. *Suppose d is a metric on $X = X_1 \times \cdots \times X_n$ satisfying the above coordinatewise convergence property. Then the open sets in (X, d) are precisely the unions of product sets*

$$U_1 \times \cdots \times U_n,$$

where each U_k is open in X_k.

Proof. First, if $E_k \subset X_k$ is closed for each k, then

$$E_1 \times \cdots \times E_n$$

is closed in X. Indeed, if $x^{(m)} \in E_1 \times \cdots \times E_n$ and $x^{(m)} \to x$ in X, then by coordinatewise convergence, $x_k^{(m)} \to x_k$ in X_k. Since each E_k is closed, $x_k \in E_k$, hence $x \in E_1 \times \cdots \times E_n$.

Now let $U_k \subset X_k$ be open. Then

$$V^{(k)} := X_1 \times \cdots \times X_{k-1} \times U_k \times X_{k+1} \times \cdots \times X_n$$

is open in X, since its complement is of the form

$$X_1 \times \cdots \times (X_k \setminus U_k) \times \cdots \times X_n,$$

which is closed by the previous remark. Hence

$$U_1 \times \cdots \times U_n = \bigcap_{k=1}^{n} V^{(k)}$$

is open in X, and any union of such products is open.

Conversely, let $U \subset X$ be open and $x = (x_1, \ldots, x_n) \in U$. We show there exist open sets $U_k \subset X_k$ such that

$$x \in U_1 \times \cdots \times U_n \subset U.$$

Suppose not. Then for each $m \in \mathbb{N}$,

$$B_{X_1}(x_1, 1/m) \times \cdots \times B_{X_n}(x_n, 1/m) \not\subset U.$$

Thus there exists $x^{(m)} \notin U$ with

$$d_k(x_k^{(m)}, x_k) < \frac{1}{m}, \qquad k = 1, \ldots, n.$$

Then $x_k^{(m)} \to x_k$ for each k, hence $x^{(m)} \to x$ in X. Since $X \setminus U$ is closed and each $x^{(m)} \in X \setminus U$, we conclude $x \in X \setminus U$, a contradiction. Therefore such open sets U_k exist, and U is a union of product sets. □

Theorem 3 (Heine–Borel Theorem)**.** *For a subspace $E \subseteq \mathbb{R}^n$, the following are equivalent:*

(i) E is compact;

(ii) every sequence in E has a convergent subsequence with limit in E;

(iii) E is closed and bounded.

Proof. By the general compactness theorem for metric spaces, for any metric space the following are equivalent: compactness, sequential compactness, and completeness together with total boundedness.

Now let $E \subseteq \mathbb{R}^n$.

If E is closed and bounded, then E is complete (since a subspace of $\mathbb{R}^n$ is complete if and only if it is closed) and totally bounded (since a subset of $\mathbb{R}^n$ is totally bounded if and only if it is bounded). Hence E is compact, and therefore every sequence in E has a convergent subsequence.

Conversely, if E is compact, then by the same general theorem E is complete and totally bounded. Since E is a subspace of $\mathbb{R}^n$, completeness implies that E is closed, and total boundedness implies that E is bounded. Thus E is closed and bounded.

Therefore (i), (ii), and (iii) are equivalent. □

Theorem 4 (Uniform Continuity Theorem). *Let (X, d) and (Y, ρ) be metric spaces, and suppose that X is compact. Then every continuous function $f : X \to Y$ is uniformly continuous.*

Proof. Suppose f is not uniformly continuous. Then there exists $\varepsilon > 0$ such that for every $m \in \mathbb{N}$ there are points $x_m, z_m \in X$ with

$$d(x_m, z_m) < \frac{1}{m} \qquad \text{but} \qquad \rho\big(f(x_m), f(z_m)\big) \geq \varepsilon.$$

Since X is compact, the sequence (x_m) has a convergent subsequence; passing to that subsequence, we may assume

$$x_m \to x \in X.$$

Because $d(x_m, z_m) \to 0$, also

$$z_m \to x.$$

By continuity of f at x,

$$f(x_m) \to f(x) \qquad \text{and} \qquad f(z_m) \to f(x).$$

Hence

$$\rho\big(f(x_m), f(z_m)\big) \leq \rho\big(f(x_m), f(x)\big) + \rho\big(f(x), f(z_m)\big) \to 0,$$

which contradicts

$$\rho\big(f(x_m), f(z_m)\big) \geq \varepsilon.$$

Therefore f is uniformly continuous. $\square$

Theorem 5 (Continuity and boundedness of linear operators). *Let X and Y be normed spaces and let $T : X \to Y$ be linear. The following are equivalent:*

(i) T is continuous;

(ii) T is continuous at 0;

(iii) T is bounded, i.e. there exists $C > 0$ such that

$$\|T(x)\| \leq C\|x\| \quad \textit{for all } x \in X.$$

Proof. (i)⇒(ii) is immediate.

(ii)⇒(iii): Since T is continuous at 0, there exists $\delta > 0$ such that

$$\|x\| \leq \delta \;\Rightarrow\; \|T(x)\| \leq 1.$$

For arbitrary $x \neq 0$, set $y = \frac{\delta}{\|x\|}x$, so $\|y\| = \delta$ and hence

$$\|T(y)\| \leq 1.$$

By linearity,

$$\|T(x)\| = \frac{\|x\|}{\delta}\|T(y)\| \leq \frac{1}{\delta}\|x\|.$$

Thus T is bounded.

(iii)⇒(i): If $\|T(x)\| \leq C\|x\|$, then for all $x, x_0 \in X$,

$$\|T(x) - T(x_0)\| = \|T(x - x_0)\| \leq C\|x - x_0\|,$$

so T is Lipschitz, hence continuous. $\square$

Theorem 6 (Principle of Uniform Boundedness). *Let X and Y be Banach spaces and let $\{T_\alpha\}_{\alpha\in A}$ be a family of continuous linear operators $T_\alpha : X \to Y$. Suppose that for each $x \in X$,*

$$\sup_{\alpha\in A} \|T_\alpha x\| < \infty.$$

Then

$$\sup_{\alpha\in A} \|T_\alpha\| < \infty.$$

Proof. For $n \in \mathbb{N}$, define

$$E_n := \left\{ x \in X : \sup_{\alpha\in A} \|T_\alpha x\| \le n \right\}.$$

Then each E_n is closed, $E_n \subseteq E_{n+1}$, and by assumption

$$X = \bigcup_{n=1}^{\infty} E_n.$$

By the Baire Category Theorem, some E_m has nonempty interior. Thus there exist $x_0 \in X$ and $\varepsilon > 0$ such that

$$B(x_0, \varepsilon) \subseteq E_m.$$

Let $c := \sup_{\alpha\in A} \|T_\alpha x_0\| < \infty$. If $\|x\| < 1$, then $x_0 + \varepsilon x \in E_m$, hence

$$\|T_\alpha(x_0 + \varepsilon x)\| \le m.$$

Thus

$$\varepsilon\|T_\alpha x\| = \|T_\alpha(x_0 + \varepsilon x) - T_\alpha(x_0)\| \le m + c.$$

Therefore

$$\|T_\alpha x\| \le \frac{m+c}{\varepsilon} \quad \text{for all } \|x\| < 1.$$

Hence $\|T_\alpha\| \le \frac{m+c}{\varepsilon}$ for all $\alpha \in A$, and so

$$\sup_{\alpha\in A} \|T_\alpha\| < \infty.$$

$\square$

Topologies

Definition 6. *Let X be a set. A* topology *on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets of X such that*

(i) $\varnothing \in \mathcal{T}$ and $X \in \mathcal{T}$,

(ii) if $\{U_\alpha\}_{\alpha\in I} \subseteq \mathcal{T}$, then

$$\bigcup_{\alpha\in I} U_\alpha \in \mathcal{T},$$

(iii) if $U_1, \dots, U_n \in \mathcal{T}$, then

$$\bigcap_{i=1}^{n} U_i \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is called a topological space. *The sets $U \in \mathcal{T}$ are called* open sets. *A subset $F \subseteq X$ is called* closed *if its complement is open.*

Separation Axioms

Definition 7 (T_0 space)**.** *A topological space $(X, \mathcal{T})$ is called a T_0-*space *if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that*

$$x \in U,\ y \notin U \quad \text{or} \quad y \in U,\ x \notin U.$$

Definition 8 (T_1 space)**.** *A topological space $(X, \mathcal{T})$ is called a T_1-*space *if for every pair of distinct points $x, y \in X$, there exists an open set $U \in \mathcal{T}$ such that*

$$y \in U \quad \text{and} \quad x \notin U.$$

Equivalently, $(X, \mathcal{T})$ is T_1 if and only if every singleton set $\{x\}$ is closed in X.

Definition 9 (T_2 space / Hausdorff)**.** *A topological space $(X, \mathcal{T})$ is called a T_2-*space*, or* Hausdorff*, if for every pair of distinct points $x, y \in X$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$x \in U, \qquad y \in V, \qquad U \cap V = \varnothing.$$

Definition 10 (Regular space)**.** *A topological space $(X, \mathcal{T})$ is called* regular *if for every closed set $F \subseteq X$ and every point $x \in X \setminus F$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$x \in U, \qquad F \subseteq V, \qquad U \cap V = \varnothing.$$

Definition 11 (T_3 space)**.** *A topological space $(X, \mathcal{T})$ is called a T_3-*space *if it is both T_1 and regular.*

Definition 12 (Normal space)**.** *A topological space $(X, \mathcal{T})$ is called* normal *if for any two disjoint closed sets $F, G \subseteq X$, there exist open sets $U, V \in \mathcal{T}$ such that*

$$F \subseteq U, \qquad G \subseteq V, \qquad U \cap V = \varnothing.$$

Definition 13 (T_4 space)**.** *A topological space $(X, \mathcal{T})$ is called a T_4-*space *if it is both T_1 and normal.*

Theorem 7. *Every compact Hausdorff space is normal. In particular, every compact Hausdorff space is a T_4-space.*

Proof. Let $F, G \subseteq X$ be disjoint closed sets. Since X is Hausdorff, for each $x \in F$ and $y \in G$ there exist disjoint open sets $U_{x,y}$ and $V_{x,y}$ such that

$$x \in U_{x,y}, \qquad y \in V_{x,y}, \qquad U_{x,y} \cap V_{x,y} = \varnothing.$$

For fixed $x \in F$, the sets $\{V_{x,y}\}_{y \in G}$ form an open cover of G. Since G is closed in the compact space X, it is compact, so we may choose a finite subcover

$$G \subseteq V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Let

$$U_x := U_{x,y_1} \cap \cdots \cap U_{x,y_n}, \qquad V_x := V_{x,y_1} \cup \cdots \cup V_{x,y_n}.$$

Then $x \in U_x$, $G \subseteq V_x$, and $U_x \cap V_x = \varnothing$.

The family $\{U_x\}_{x \in F}$ is an open cover of F. Since F is compact, choose a finite subcover

$$F \subseteq U_{x_1} \cup \cdots \cup U_{x_m}.$$

Define

$$U := U_{x_1} \cup \cdots \cup U_{x_m}, \qquad V := V_{x_1} \cap \cdots \cap V_{x_m}.$$

Then U and V are open, $F \subseteq U$, $G \subseteq V$, and $U \cap V = \varnothing$. Hence X is normal. $\square$

Remark 2 (Continuity criterion in metric spaces). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. A function $f : X \to Y$ is continuous at $x \in X$ if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that*

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Equivalently, $f(U) \subseteq B(f(x), \varepsilon)$.

Theorem 8 (Uniform limit preserves continuity). *Let $(X, \mathcal{T})$ be a topological space and (Y, d) a metric space. Suppose (f_n) is a sequence of functions $f_n : X \to Y$ converging uniformly to $f : X \to Y$. If each f_n is continuous at $x \in X$, then f is continuous at x.*

Proof. We first recall the following characterization of continuity at a point: f is continuous at x if and only if for every $\varepsilon > 0$ there exists a neighbourhood U of x such that

$$d(f(y), f(x)) < \varepsilon \quad \text{for all } y \in U.$$

Now let $\varepsilon > 0$. By uniform convergence, there exists $N \in \mathbb{N}$ such that

$$d(f_N(y), f(y)) < \frac{\varepsilon}{3} \quad \text{for all } y \in X.$$

Since f_N is continuous at x, there exists a neighbourhood U of x such that

$$d(f_N(y), f_N(x)) < \frac{\varepsilon}{3} \quad \text{for all } y \in U.$$

Then for all $y \in U$,

$$d(f(y), f(x)) \leq d(f(y), f_N(y)) + d(f_N(y), f_N(x)) + d(f_N(x), f(x)) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon.$$

By the above characterization, f is continuous at x. □

Basis & Subbasis

Definition 14. *Let X be a set. A collection $\mathcal{B}$ of subsets of X is called a* basis *for a topology on X if:*

(i) For every $x \in X$ there exists $B \in \mathcal{B}$ such that $x \in B$.

(ii) If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \mathcal{B}$, then there exists $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Theorem 9 (Topology generated by a basis). *If $\mathcal{B}$ satisfies the two conditions above, then there exists a unique topology $\mathcal{T}$ on X whose open sets are precisely the unions of elements of $\mathcal{B}$. That is,*

$$\mathcal{T} = \left\{ \bigcup_{\alpha \in I} B_\alpha : B_\alpha \in \mathcal{B},\ I \text{ arbitrary} \right\}.$$

Definition 15. *Let X be a set. A collection $\mathcal{S}$ of subsets of X is called a* subbasis *for a topology on X if the union of all sets in $\mathcal{S}$ equals X,*

$$X = \bigcup_{S \in \mathcal{S}} S.$$

The topology generated by $\mathcal{S}$ is obtained by first forming the basis consisting of all finite intersections of elements of $\mathcal{S}$,

$$\mathcal{B} = \{S_1 \cap \cdots \cap S_n : S_i \in \mathcal{S},\ n \in \mathbb{N}\},$$

and then taking arbitrary unions of these sets.

The topology generated by the subbasis $\mathcal{S}$ consists of all arbitrary unions of elements of this basis. That is, a set $U \subseteq X$ is open if and only if it can be written as

$$U = \bigcup_{\alpha \in I} \left(\bigcap_{i=1}^{n_\alpha} S_{\alpha,i} \right), \qquad S_{\alpha,i} \in \mathcal{S}, \quad n_\alpha \in \mathbb{N}.$$